

RANSAC

RANSAC

- Random sample consensus
- It is an iterative method for estimating a mathematical model from a data set that contains outliers.
- The RANSAC algorithm works by identifying the outliers in a data set
- It estimates the desired model using data that does not contain outliers.

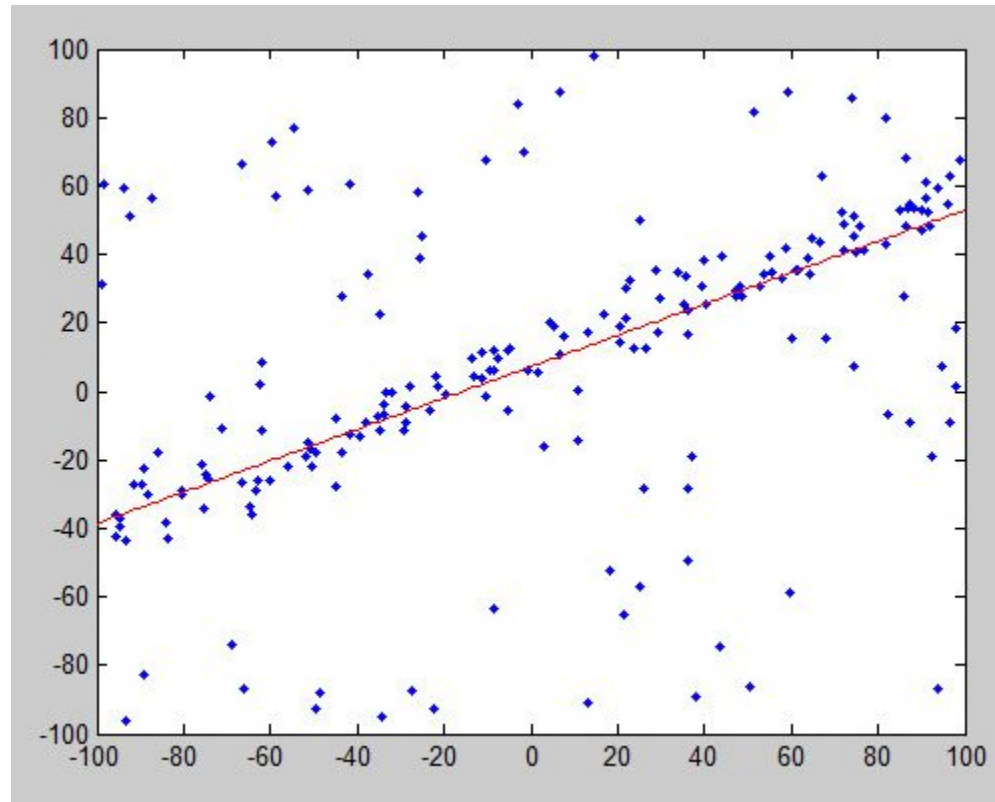
It is widely used in:

- Line fitting
- Homography estimation
- Fundamental matrix estimation
- Pose estimation (PnP)
- 3D plane fitting

Steps

- Randomly selecting a subset of the data set
- Fitting a model to the selected subset
- Determining the number of outliers
- Repeating steps 1-3 for a prescribed number of iterations

Data points shown in blue, with the line of form $y = mx + c$ estimated using RANSAC indicated in red



- In computer vision, RANSAC is used as a robust approach to
 1. It estimates the fundamental matrix in stereo vision
 2. It finds the commonality between two sets of points for feature-based object detection, and
 3. It registers sequential video frames for video stabilization.

Example:-

illustrate the process of applying a 3×3 kernel to a 3×3 image using numerical examples.

Step 1: Image Matrix

$$I = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Step 2: Kernel Matrix

Example edge detection kernel

$$K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

Step 3: Convolution Operation

Multiply corresponding elements and sum:

$$\begin{aligned} & (1 \times 1) + (2 \times 0) + (3 \times -1) \\ & + (4 \times 1) + (5 \times 0) + (6 \times -1) \\ & + (7 \times 1) + (8 \times 0) + (9 \times -1) \end{aligned}$$

Step 4: Calculate

$$\begin{aligned} &= 1 + 0 - 3 \\ &+ 4 + 0 - 6 \\ &+ 7 + 0 - 9 \\ &= -6 \end{aligned}$$

So the output pixel value = -6.